

Aman Malik

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Profile

Computer Science undergraduate with hands-on experience in machine learning, deep learning, computer vision, NLP, and agentic AI systems. Experienced in developing production-grade AI agents, RAG applications, and intelligent systems from scratch, with a strong interests in AI research and real-world intelligent systems.

Education

Bachelor of Science in Computer Science 08/2023 – Present
FAST NUCES
Karachi, Pakistan
CGPA: 3.73

A levels 08/2021 – 05/2023
Highbrow School & College
Karachi, Pakistan
• Mathematic: A
• Physics: A
• Chemistry: B

Technical Skills

Languages Python, C/C++, SQL, PostgreSQL	Generative AI & Agentic Systems LLMs, RAG, Vector Databases, Agentic AI, Tool Calling, Multi-Agent Systems, n8n
AI/ML & Frameworks PyTorch, TensorFlow/Keras, Scikit-learn, Hugging Face Transformers, LangChain, OpenCV, NumPy, Matplotlib	Domains Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, Conversational AI, Recommender Systems

Professional Experience

Agentic AI Engineer 04/2026 – Present
BizBuddy
Karachi, Pakistan
• Built and improved production-grade agentic AI systems for real-world business workflows using LLMs, n8n, APIs, and database integrations.
• Independently developed AI agents from scratch, including tool-calling, workflow logic, and reliability safeguards.
• Debugged and optimized existing agents to improve accuracy, robustness, and end-to-end workflow reliability.

Publications

Beyond Prompt Engineering: A Controlled Evaluation of Deterministic Guardrails in a Transactional LLM Agent 2026
Zenodo
• Conducted a controlled evaluation of **prompt-only vs. tool-grounded LLM agents across 20 transactional scenarios**, achieving **70% vs. 90% task success**.
• Analyzed **deterministic guardrails, tool grounding, routing/orchestration failures, and safe transaction validation** in real-world LLM agent workflows.
• Published as a **research preprint on Zenodo** with an accompanying technical article on Medium.

Projects

AI-Powered Multilingual Translator
Python | Hugging Face Transformers | Meta NLLB-200 | Streamlit
• Developed an AI-powered multilingual translation application using Meta NLLB-200, enabling translation across 200+ languages with an interactive Streamlit interface.
• Implemented transformer-based neural machine translation with Hugging Face Transformers and optimized inference using model caching for efficient performance.

Face Emotion Detection System
CNN, TensorFlow/Keras, OpenCV
• Designed and trained a custom CNN from scratch on the FER2013 dataset (35K+ images) to classify facial expressions into 7 emotion categories, achieving 67.6% test accuracy using data augmentation, batch normalization, and dropout regularization
• Built a real-time emotion detection pipeline using OpenCV for face detection, integrated with the trained model to support both live webcam and static image inference

Reinforcement Learning Navigation Agent
Python | NumPy | Q-Learning | Tkinter
• Implemented a Q-Learning navigation agent from scratch using NumPy, with epsilon-greedy exploration, Bellman updates, and reward shaping in dynamic grid environments.
• Built a real-time GUI to visualize training, learned policies, Q-value heatmaps, and agent navigation across configurable environments.